

Exponential Equations Not Requiring Logarithms

Date_____ Period____

Solve each equation.

1) $5^{3n} = 125$

2) $2^{2n} = 16$

3) $5^{3r} = 5^{-2r}$

4) $3^{-2k} = 81$

5) $2^{-3x} = 2^{x-1}$

6) $6^{3m} = 36$

7) $10^{3x} = 10000$

8) $4^{r+1} = 1$

9) $\left(\frac{1}{8}\right)^{3x} \cdot 64^{2x+1} = 4$

10) $32^{2x} = 8$

11) $6^{-3v-2} = 36$

12) $243^x = 81$